

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js automation service** for MCA filings, reducing manual filing time by **85%** with retries, browser-state recovery, and API-backed validation.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized backend services with **Docker**, deployed to **Google Cloud Run**, and automated CI/CD with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, integrating model inference, preprocessing, and visualization into an existing desktop medical-imaging application.
- Reduced processing time by **87%** using asynchronous and parallel execution across expensive image-processing workflows.
- Improved 3D reconstructed surface visualization while preserving compatibility with existing VTK/wxPython user flows.

Open Source

[Contributions](#) 

- Contributed to Invesalio, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, adapting implementation details to established open-source review and release practices.

Projects

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#) 

- Built containerized Go/Rod services for a **content distribution platform** using browser automation, with async workers, RabbitMQ retries, Redis state, and PostgreSQL persistence.
- Designed the system for operational simplicity, service isolation, and recoverable execution around unreliable external integrations and long-running jobs.

Seaweed: Contest Platform | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

[GitHub](#) 

- Built backend services for registration, submissions, sandboxed execution, scoring, and live contest rankings across **500+ concurrent users**.
- Used async Go workers, Judge0, PostgreSQL, Redis, and Docker/Kubernetes deployment paths on AWS to keep evaluation workflows recoverable under contest load.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built a gateway that preserves context around prompts, tools, retrieved data, policy decisions, and outputs, making AI-agent actions inspectable through structured traces and audit ledgers.
- Modeled runtime settings, route profiles, RBAC decisions, and attack-lab outcomes as explicit metadata so agent behavior could be governed, replayed, and debugged.

Technical Skills

Languages: Go, Python, TypeScript

Frameworks: Echo, FastAPI, Gin, Node.js

Tools: Docker, Google Cloud Run, GitHub Actions, PostgreSQL, Redis, RabbitMQ, Prometheus, AWS